

Ege Özgür

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Based in Istanbul, Türkiye — Available for full-time work

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SKILLS

- **Programming Languages:** C++, C#, Java, JavaScript, Python, PHP, SQL, Verilog
- **Game Development:** Unreal Engine 5, Blueprints, Behavior Trees, Blackboards, NavMesh, Niagara, UMG, DirectX 12, Gameplay Programming, Level Design, Computer Graphics
- **Frameworks & Tools:** Git/GitHub, Visual Studio, Visual Studio Code, .NET MVC, Node.js, Spring, Android Studio, Autodesk Fusion
- **Databases:** PostgreSQL, MySQL
- **Methods:** Agile Development, Scrum, Software Testing, Technical Documentation
- **Languages:** Turkish (Native), English (C1)

EDUCATION

- **The University of Warwick** Coventry, United Kingdom
MSc Games Engineering Sep. 2025 – Sep. 2026
- **Sabancı University** Istanbul, Turkey
BSc Computer Science and Engineering; GPA: 2.96 / 4.00 Sep. 2021 – Jun. 2025

EXPERIENCE

- **Farmazon** Istanbul, Turkey
Software Engineer Intern – Mandatory Summer Internship Jul. 2024 – Sep. 2024
 - **Feature Design and Implementation:** Independently designed and built a GitHub Pull Request Checker tool from concept to delivery using .NET MVC, defining the data flow, interface layout, and filtering logic before implementation. Integrated the GitHub API and restructured JSON responses to produce a clean, readable display informed by how developers actually scan pull request data.
 - **Cross-Functional Collaboration:** Worked within an Agile team, documenting feature specifications, presenting iterations for feedback, and adjusting design decisions based on stakeholder input — championing the feature from inception to live delivery.
- **Makswin** Istanbul, Turkey
Software and Application Tester – Voluntary Internship Jul. 2023 – Aug. 2023
 - **Test Planning and Quality Assurance:** Wrote and executed detailed test plans for software and mobile applications, identifying edge cases, documenting reproduction steps, and providing structured feedback to development teams. Developed a methodical, detail-oriented approach to quality validation across multiple feature areas.
 - **Cross-Team Collaboration:** Operated within a fast-paced corporate environment, coordinating with engineering and product teams to communicate findings clearly and support feature readiness ahead of release.

PROJECTS

- **MSc Dissertation – Visual Analytics for Debugging and Optimising Rendering Performance in Real-Time Game Development (September 2026):** Designed and implemented a controlled Unreal Engine 5.4 benchmark environment and a web-based visual analytics tool for rendering-performance analysis. Collected and compared frame-time, CPU/GPU, hitch-rate, and pass-level telemetry across repeated runs using statistical confidence intervals. Evaluated performance–quality trade-offs across LOD, shadow quality, instancing, resolution scaling, and post-processing configurations.
- **Phoenix Ascendant – Unreal Engine 5 Action Game (June 2026):** Developed a two-level third-person action game in Unreal Engine 5.4.4 using C++ and Blueprints. Implemented elemental combat, reusable health and damage systems, wave-based enemy AI, UMG interfaces, Niagara effects, spatial audio, and a complete menu-to-victory game flow.

- **Skyfire Uprising – Level Designer, Game Engine Design and Development (May 2026):** Designed and implemented a spaceport and research-facility FPS level in Unreal Engine 5, focusing on player flow, pacing, landmarks, traversal routes, encounter placement, and modular environment planning within a six-person Scrum team.
- **Physically Based Renderer – Advanced Computer Graphics (May 2026):** Extended a C++ path tracer with light tracing, instant radiosity, multiple BSDFs, multiple importance sampling, environment-map sampling, SAH BVH acceleration, multithreading, and Intel Open Image Denoise. Achieved a 12.5x render-stage speedup. Project available on GitHub: github.com/ege-ozgur/RTBase-Renderer
- **Critical Review: Nanite vs. Traditional LOD Techniques (April 2026):** Authored an academic review comparing Nanite with traditional LOD techniques, focusing on performance, workflow, limitations, fallback meshes, and hybrid geometry pipelines.
- **Tactical Five – Football Strategy Game (January 2026):** Designed a football strategy game with card-based actions, possession mechanics, probability-driven outcomes, and MDA-based balancing.
- **3D Game Development with DirectX 12 (December 2025):** Built a custom 3D game in C++ and DirectX 12 with PSO management, constant buffers, shader reflection, instancing, skeletal animation, camera controls, and real-time rendering. Project available on GitHub: github.com/ege-ozgur/3D-Game-Directx12
- **3D Survivor Game (C++) (November 2025):** Developed a survivor-style game in C++ with player movement, enemy AI, collision detection, camera control, and a real-time update loop. Project available on GitHub: github.com/ege-ozgur/3D-SurvivorGame
- **Central LA Application System (Sep. 2024 – Jun. 2025):** Built a university application-management system in a five-person team, contributing to requirements analysis, PHP backend architecture, and iterative feature development.
- **GitHub Pull Request Checker App (Jul. 2024 – Sep. 2024):** Created a .NET MVC web tool that retrieves, filters, and presents GitHub pull-request data through the GitHub API.
- **Flight Management App (Feb. 2024 – Jun. 2024):** Designed the domain model and PostgreSQL schema, then implemented backend logic and APIs with Node.js.
- **FUTCARDS Mobile App (Sep. 2023 – Jan. 2024):** Developed a FIFA-style player-card application with a Java/Spring backend and Android front end.
- **Air Hockey Game on FPGA (Sep. 2023 – Jan. 2024):** Designed ASM charts, implemented game logic in Verilog, created test benches, and deployed the game on an FPGA using Vivado.
- **Natural Disasters Database Project (Feb. 2023 – Jun. 2023):** Built a MySQL database with views, procedures, and triggers, then produced visual analytics with Python and matplotlib.

INTERESTS

- **Football:** Playing, watching, and playing Football Manager and EA FC
- **Collecting:** Football jerseys, stickers, trading cards, and magnets
- **Other:** Video Games, Travelling